Fig. 12: Honeywell's ADIRS

Fig. 13: Sigma-95N

freedom. Along with accelerometers, RLGs constitute the INS. Fig. 9 shows one such RLG-based INS package. The system is a high-grade IMU that meets the accuracy requirements of attitude, orientation, position and navigation. The system is characterised by bias stability of 0.1°/hr, scale factor stability of 75PPM and ARW specification equal to $0.028^\circ/\sqrt{\text{hr}}$.

HG9900U from Honeywell Aerospace (Fig. 10) is another navigation-grade IMU. It mainly comprises Honeywell GG1320AN digital laser gyroscopes, Honeywell QA2000 accelerometers and associated electronics.

Laser gyroscopes in this INS package are characterised by one sigma error coefficient of $<0.003^\circ$/hr (bias), $<0.002^\circ/\sqrt{\text{hr}}$ (ARW) and <5ppm (scale factor). Accelerometers are characterised by one sigma error coefficient of $<25\mu$g (bias) and <100ppm (scale factor).

HG1700 (Fig. 11) from Honeywell Aerospace is another high-performance, tactical-grade IMU designed for a wide range of guidance and control applications. The IMU package comprises three RLGs, three quartz resonating beam accelerometers and associated electronics, environmentally sealed in rugged aluminium housing. The IMU is available in four different performance grades with one sigma gyro bias of 1°/hr, 2°/hr, 3°/hr and 5°/hr. Corresponding maximum random walk specifications are $0.125^\circ/\sqrt{\text{hr}}$, $1^\circ/\sqrt{\text{hr}}$, $2^\circ/\sqrt{\text{hr}}$ and $3^\circ/\sqrt{\text{hr}}$, respectively.

Yet another inertial navigation package from Honeywell, popular with a wide range of passenger and military aircraft, is air data inertial reference system (ADIRS). It is configured around HG-2030 and HG-2050 series air data inertial reference units (ADIRUs). Fig. 12 shows the ADIRS. The ADIRU uses digital RLGs, which, along with quartz accelerometers, provide position and attitude information.

Sigma-95 from SAGEM, France is a high-performance navigation system. It has two variants, namely, Sigma-95L and Sigma-95N. Sigma 95L is a lightweight and compact INS designed for fixed and rotary wing aircraft, offering an excellent performance level. It is particularly adapted to Male unmanned aerial vehicles (UAVs) and pod applications.

With an integrated SAGEM-designed GPS receiver, Sigma 95L can be used either as an autonomous attitude and heading reference system or as a full-size INS. Sigma 95L offers different levels of performance and several modes of alignment to adapt to the different operational needs of users.

Sigma-95N (Fig. 13) is designed for the most demanding applications requiring high navigation and guidance accuracy. It can integrate all available sensors including GPS, Glonass, air data system and so on, in all types of avionic configurations. In pure inertial mode, it offers position accuracy of <0.5 Nm/hr, velocity accuracy of <0.7 m/s, pitch and roll accuracy of $<0.03^\circ$, and heading accuracy of $<0.05^\circ$. In hybrid mode, it offers position accuracy of <10m.

GPS ADIRS from SAGEM is another RLG-based navigation package. Sigma-40 inertial navigation system from SAGEM with heading accuracy of $<0.05^\circ$, and pitch and roll accuracy of $<0.02^\circ$, and position accuracy of <1.5Nm/24 hours in autonomous navigation mode is an advanced inertial attitude and heading reference system that employs laser gyroscopic technology. It is primarily designed for maritime applications meeting the most demanding navigation and weapon system stabilisation requirements. It is one of the most widely-used naval inertial navigation systems in use on both surface vessels and submarines.

Northrop Grumman's LTN-101 Flagship integrated global air data inertial navigation unit (GNADIRU) is another major inertial navigation system offering position accuracy of 2.0Nm/hr in pure inertial mode. The inertial sensor in the package combines two 18cm Zero-Lock laser gyroscopes in a single cavity.